

Exercises

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Surname, First name**5EZB0 Math 2**

Final exam Q4

1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9
0	0	0	0	0	0	0

Particular Ans on paper exam instructions

- Write in a black or blue pen.
- You answer open-ended questions by using the text box. Provide your answers on the papers inside the answer box underneath a question. **If you need more space for your answers, use the extra space at the end of the exam, and clearly indicate there which question you continue answering. In the text box of the particular question, clearly state that you proceed with your answer on a different page.**
- **The solutions should be formulated clearly, arranged orderly, and should include clear explanations of all steps that are needed to obtain the final answer.**
- Hand in all pages. Do not remove the staple. If you remove it anyhow, check that you hand in all pages.

Dear student,

This is the final exam for 5EZB0. Write down your name and your student ID at the appropriate places above. Make sure that you enter your student ID by fully coloring the appropriate boxes. On the examination attendance card, you fill in the PDF number. You can find the correct number on the top of the first page of your exam (e.g. 1234.pdf).

Please read the following information carefully:

Date exam: 23/06/2025

Start time: 9:00

End time: 12:00 (+30 minutes for time extension students)

Number of questions: 5

Maximum number of points/distribution of points over questions: 80=20+20+20+10+10

Contribution to the final grade = (number of points)%

Permitted examination aids

- Scrap paper (fully blank)
- Calculator(**non-graphical/ non-programmable**)

Important:

- You are only permitted to visit the toilets under supervision
- Examination scripts (fully completed examination paper, stating name, student number, etc.) must always be handed in
- The house rules must be observed during the examination
- The instructions of subject experts and invigilators must be followed
- Keep your work place as clean as possible: put pencil case and breadbox away, limit snacks and drinks
- You are not permitted to share examination aids or lend them to each other
- Do not communicate with any other person by any means

During written examinations, the following actions will in any case be deemed to constitute fraud or attempted fraud:

- using another person's proof of identity/campus card (student identity card)
- having a mobile telephone or any other type of media-carrying device on your desk or in your clothes
- using, or attempting to use, unauthorized resources and aids, such as the internet, a mobile telephone, smartwatch, smart glasses etc.
- having any paper at hand other than that provided by TU/e, unless stated otherwise
- copying (in any form)
- visiting the toilet (or going outside) without permission or supervision

You can start the exam now, good luck!



Consider the function

$$f(x, y) = \sqrt{1 + y^2 - x^2}$$

3p **1a** What is the maximal domain of definition $\mathcal{D} \subseteq \mathbb{R}^2$ of the function f ? Sketch the domain with explanation.

[illegible]

2p

[illegible]

3p

[illegible]



3p **1d** Is the function f continuous in $\mathcal{D}$? Justify briefly.

In particular, find the value of

$$\lim_{(x,y) \rightarrow (1,0)} f(x,y)$$



4p

Does $\lim_{(x,y) \rightarrow (1,0)} \partial_x f(x,y)$ exist? Briefly state the reason.

[illegible]

3p **1f** Along which unit vector $\hat{u}$ is the function $f(x, y)$ increasing the fastest at $(1, 1)$?

2p **1g** Let $L(x, y)$ denote the linear approximation of the function f around the point $(1, 1)$. Determine $L(1, 0)$, i.e., the value of this linearization at the point $(1, 0)$.



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$$W = \int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r}$$

2a

2a

[illegible]

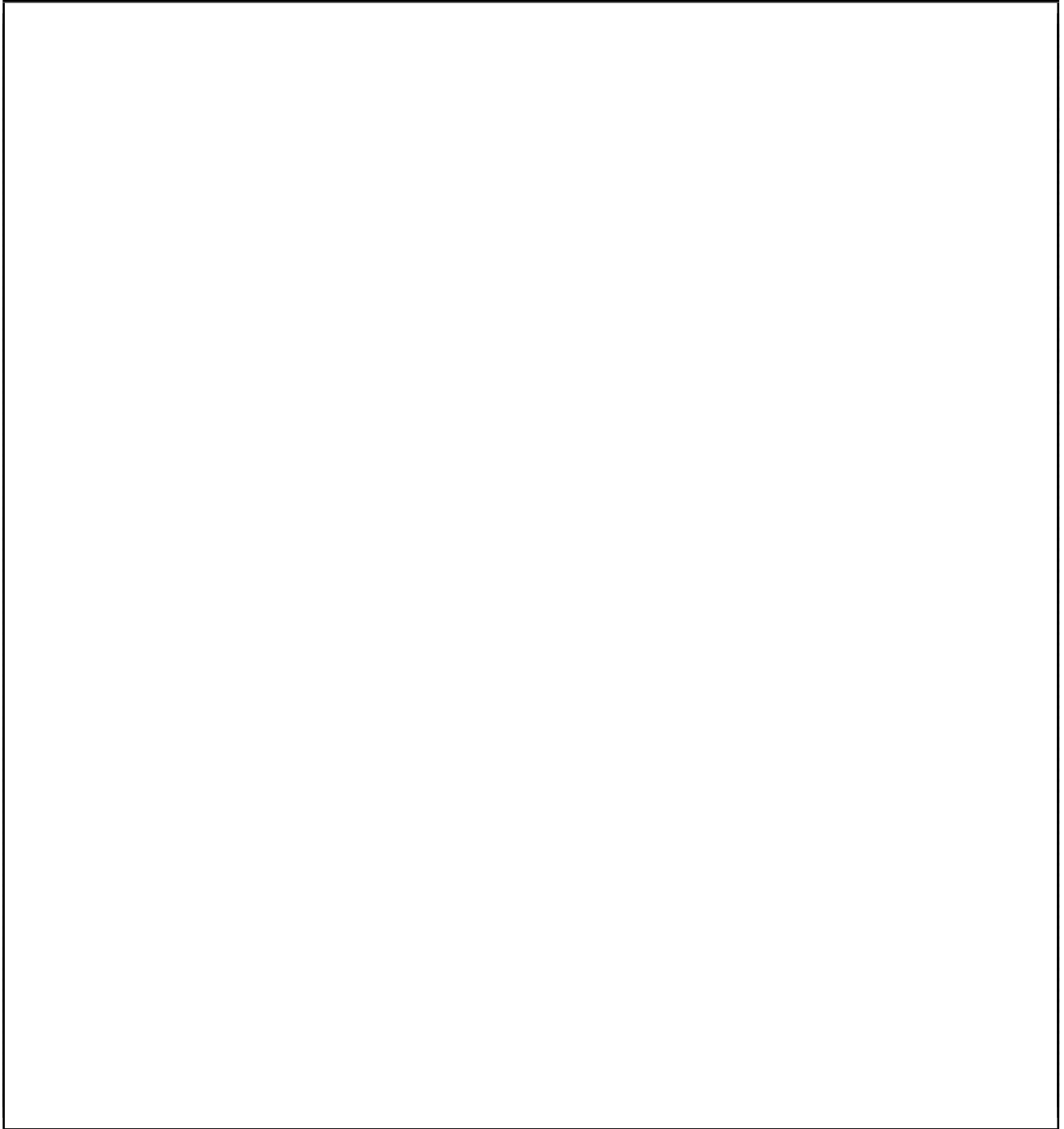
5p **2b** Show that $\mathbf{F}$ is a conservative vector field. Determine the potential U such that $\mathbf{F} = -\nabla U$.



- 3p **2c** Now consider the work done along a straight line $\mathcal{C}_2$ from the origin to the same endpoint $(3, \sqrt{3})$. Calculate or determine the work $W = \int_{\mathcal{C}_2} \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r}$.



- 5p **2d** Find the streamline of $\mathbf{F}$ passing through the point (x_0, y_0) . Draw the streamlines in the xy -plane.



Qu

8p

$$\int_0^1 \int_{\sqrt{y}}^1 \frac{e^{x^2}}{x} dx dy$$



12p **3b**

Let $\mathcal{D} := \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 4, x^2 + y^2 \leq 2x, y \geq 0\}$. Determine $\iiint_{\mathcal{D}} \frac{xy}{x^2 + y^2} dx dy dz$.

[Hint: Use a transformation to the cylindrical coordinate system. The domain $\mathcal{D}$ is shown after the answer box.]



[illegible]

[illegible]

4 Flux through a surface $\iint_S \mathbf{f} \cdot d\mathbf{S}$: Find the flux of $\mathbf{f}(x, y, z) = \frac{-y\mathbf{i} + x\mathbf{j}}{\sqrt{x^2 + y^2}} + e^{-z^3}\mathbf{k}$ upward through the Helicoid surface S parameterized by

where $0 \leq v \leq 2\pi$ and $0 \leq u \leq v$.

[illegible]

Question 5

5p **5a** Let $\mathbf{w} = \hat{\mathbf{i}} - \hat{\mathbf{j}}$, the position vector $\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$, and its length $|\mathbf{r}| = (x^2 + y^2 + z^2)^{\frac{1}{2}}$.

Find the value of $\nabla \times (\mathbf{w} \times \mathbf{r}) - \nabla(\mathbf{w} \cdot \mathbf{r})$



5p

$$\nabla \cdot \nabla \left(\frac{1}{|\mathbf{r}|} \right) = 0$$

[illegible]

6 You can write your answers in this extra space here if you exceed the allotted space

[illegible]

